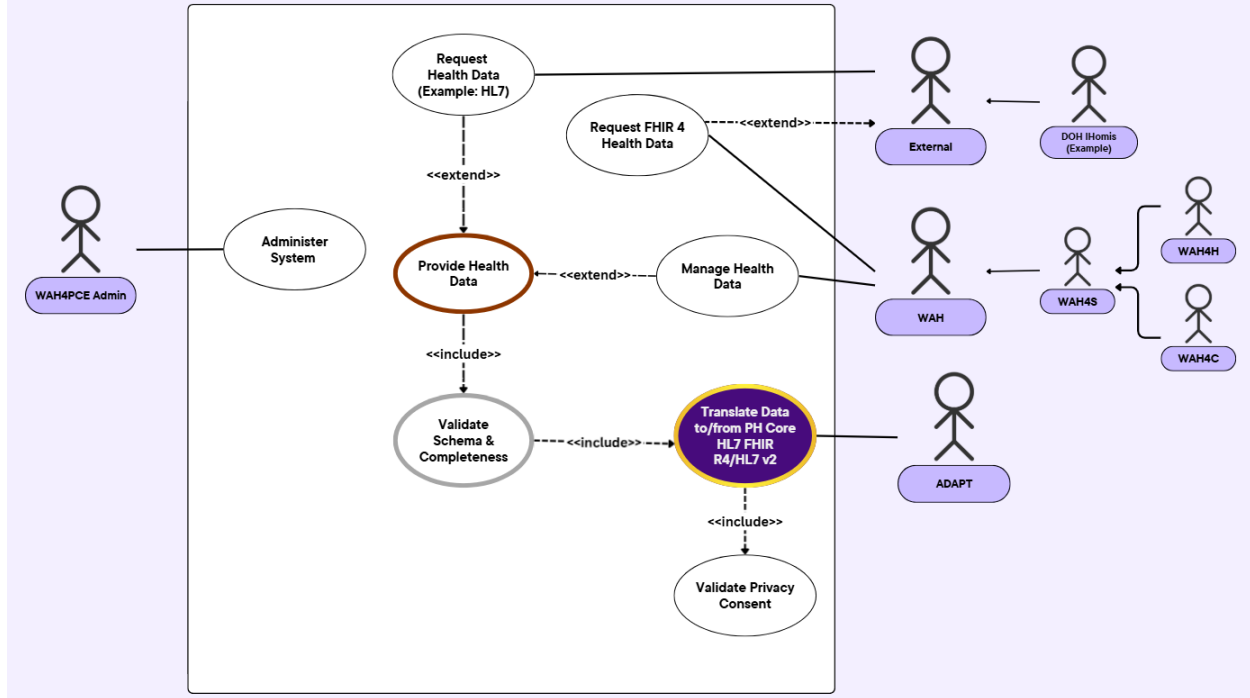


WAH4PCE INTEROPERABILITY



Use Case Name:	Administer System
Scenario:	The system administrator accesses the interoperability platform to configure endpoints, manage user credentials, and monitor system health.
Triggering Event:	The WAH4PCE Admin navigates to the administrative portal and logs in.
Brief Description:	This use case governs the administrative oversight of the platform, allowing the admin to ensure network uptime, manage security, and configure system parameters.
Actors:	Primary: WAH4PCE Admin
Related Use Cases:	None
Stakeholders:	WAH4PCE Admin: Needs secure, comprehensive access to system settings. All Users: Rely on the admin to keep the system operational and secure.
Preconditions:	The WAH4PCE Admin must be authenticated with valid administrative privileges.
Postconditions:	System configurations are successfully updated, applied, and logged in the immutable audit trail.
Flow of Activities:	Actor: WAH4PCE Admin logs into the system dashboard and selects a configuration module. System: Authenticates the Admin and presents the requested administrative interface. Actor: Admin inputs changes or acknowledges system alerts. System: Commits the changes, updates the system state, and logs the activity.
Exception Conditions:	E1: Invalid Credentials. If the Admin provides incorrect login details, the System denies access and logs a failed attempt.

Use Case Name:	Provide Health Data
Scenario:	A core data ingestion or routing event occurs on the platform, acting as the base transaction engine that can optionally be extended by requests, management, or translation processes.
Triggering Event:	Triggered implicitly when an actor initiates a data transaction (either via the Request or Manage extension points).
Brief Description:	This is the foundational data handling use case. It represents the core pipeline for routing health data within the system, automatically including translation protocols when necessary, and serving as the base that other use cases extend.
Actors:	Primary: System (External, WAH)
Related Use Cases:	Request Health Data (Example: HL7) (<<extend>>) Manage Health Data (<<extend>>) Translate Data to/from PH Core HL7 FHIR R4/HL7 v2 (<<include>>)
Stakeholders:	External / WAH Actors: Depend on this core pipeline to route data securely. Patients: Depend on this pipeline to handle their data safely.
Preconditions:	A data transaction must be initiated by an authorized external or WAH actor.
Postconditions:	The data payload is successfully processed, translated (if necessary), and routed to its appropriate destination.
Flow of Activities:	Actor: (Triggered by extending use cases). System: Initializes the core data pipeline and evaluates the incoming payload. System: Hits the <<include>> point to trigger the "Translate Data..." use case to ensure standard formatting. System: Concludes the transaction and finalizes the routing of the health data.

Exception Conditions:	E1: Pipeline Failure. If the core system experiences a critical error, the transaction is halted, and an alert is sent to the WAH4PCE Admin.

Use Case Name:	Validate Schema & Completeness
Scenario:	Upon receiving a data payload in the core pipeline, the system automatically checks the structural integrity of the data and ensures all mandatory fields are present before it can be processed or routed.
Triggering Event:	Automatically and mandatorily triggered by the "Provide Health Data" use case.
Brief Description:	An internal validation protocol designed to ensure data quality. It checks if the incoming data meets required schemas and automatically triggers translation protocols if the data needs structural normalization.
Actors:	Primary: System (Internal)
Related Use Cases:	Provide Health Data (<<include>>) Translate Data to/from PH Core HL7 FHIR R4/HL7 v2 (<<include>>)
Stakeholders:	All System Users: Rely on this process to prevent corrupted or incomplete data from entering the repository.
Preconditions:	A data payload must be active in the core processing pipeline.
Postconditions:	The data is confirmed as structurally valid and complete, allowing the transaction to continue.
Flow of Activities:	System: Intercepts the payload from the Provide Health Data pipeline. System: Scans the payload against defined structural rules and required fields. System: Hits the <<include>> point to trigger the "Translate Data..." use case to map the data to PH Core standards if needed. System: Returns a validation success signal back to the core pipeline.
Exception Conditions:	E1: Validation Failure. If the payload is malformed or missing mandatory fields, the System drops the transfer, logs the error, and returns a 400 Bad Request to the initiating actor.

Use Case Name:	Request Health Data (Example: HL7)
Scenario:	An external organization searches the interoperable repository to retrieve specific patient records in legacy or standard HL7 format.
Triggering Event:	An External actor (e.g., DOH iHomis) submits a data query or retrieval request to the platform.
Brief Description:	This use case extends the core data pipeline to allow authorized external entities to pull interoperable health data from the repository in HL7 formats.
Actors:	Primary: External (Generalized actor including DOH iHomis)
Related Use Cases:	Provide Health Data (<<extend>>)
Stakeholders:	External Actors: Need accurate and timely access to patient records.
Preconditions:	The External actor must be authenticated and authorized to access the requested data scope.
Postconditions:	The requested HL7 data is securely delivered to the actor, and the access event is logged.
Flow of Activities:	Actor: External actor submits a data request query via the integration endpoint. System: Verifies identity and authorization scope. System: Executes the core "Provide Health Data" pipeline to retrieve and route the data. System: Transmits the requested data back to the Actor.
Exception Conditions:	E1: Data Not Found. If no records match, the System returns a '404 Not Found' status.

Use Case Name:	Request FHIR 4 Health Data
Scenario:	A WAH-affiliated entity requests modern, highly structured FHIR R4 clinical data from the interoperability platform.
Triggering Event:	A WAH actor submits an API request specifically for FHIR R4 formatted data.
Brief Description:	This use case governs the secure querying and extraction of modern FHIR R4 health data tailored for actors within the WAH ecosystem, with potential extended access to External actors.
Actors:	Primary: WAH (Generalized actor including WAH4H, WAH4S, WAH4C), External
Related Use Cases:	None directly valid as an extension to UCs; extends actor access.
Stakeholders:	WAH Actors: Need structured, modern data schemas for their specific clinical applications.
Preconditions:	The actor must be authenticated and the requested data must exist in or be translatable to FHIR 4.
Postconditions:	The requested FHIR 4 data is securely delivered to the actor.
Flow of Activities:	Actor: WAH actor submits a query for FHIR R4 resources. System: Authenticates the actor and verifies access permissions. System: Queries the repository for the FHIR resources. System: Encrypts the payload and transmits it back to the Actor.
Exception Conditions:	E1: Unauthorized Access. If the actor requests resources outside their scope, the system denies the request and logs a security exception.

Use Case Name:	Manage Health Data
Scenario:	An authorized data owner updates, archives, or amends existing health records they previously submitted to the platform.
Triggering Event:	A WAH actor sends an update or modification payload for a specific existing record.
Brief Description:	This use case extends the core data pipeline to allow primary WAH actors to maintain the accuracy and lifecycle of their submitted data, ensuring the ecosystem contains the most current clinical information.
Actors:	Primary: WAH (Generalized actor including WAH4H, WAH4S, WAH4C)
Related Use Cases:	Provide Health Data (<<extend>>)
Stakeholders:	WAH Actors: Require the ability to amend patient information as clinical situations evolve.
Preconditions:	The target record must exist, and the WAH actor must be the authoritative owner of that record.
Postconditions:	The record is successfully updated (with version history maintained), and an audit trail is logged.
Flow of Activities:	Actor: WAH actor submits an update payload for an existing health record. System: Validates the Actor's authority to modify this specific data asset. System: Triggers the core "Provide Health Data" pipeline to process the payload. System: Applies the updates, versions the record, and sends a success confirmation.
Exception Conditions:	E1: Record Locked. If the record is currently in use or locked by another process, the system rejects the update and prompts a retry.

Use Case Name:	Translate Data to/from PH Core HL7 FHIR R4/HL7 v2
Scenario:	Data flowing through the core pipeline requires structural normalization to ensure it matches the PH Core HL7 FHIR R4 or HL7 v2 standards.
Triggering Event:	Mandatorily triggered by the "Provide Health Data" use case when data ingestion or extraction occurs.
Brief Description:	This included use case utilizes an AI agent to perform complex semantic mapping and translation of health data schemas, ensuring true interoperability across the platform while enforcing a mandatory privacy check.
Actors:	Primary: WAH4PCE AI
Related Use Cases:	Provide Health Data (<<include>>) Validate Privacy Consent (<<include>>)
Stakeholders:	WAH4PCE AI: Requires parameters to map data schemas accurately. All System Users: Depend on this process for standardized data.
Preconditions:	A data payload must be staged in memory, and the AI translation service must be operational.
Postconditions:	The data payload is successfully mapped to standard formats without losing clinical meaning.
Flow of Activities:	System: Routes the data payload to the AI agent during the core transaction. Actor: WAH4PCE AI analyzes the schema and performs semantic mapping. System: Pauses translation to trigger the mandatory "Validate Privacy Consent" use case. Actor: WAH4PCE AI finalizes the standardized output and returns it to the pipeline.
Exception Conditions:	E1: Translation Failure. If the AI cannot safely map the data without altering clinical meaning, it halts, quarantines the payload, and alerts the admin.

Use Case Name:	Validate Privacy Consent
Scenario:	Before translation and processing can finalize, the system checks the patient's active consent directives to ensure data sharing is legally permitted.
Triggering Event:	Automatically and mandatorily triggered by the "Translate Data..." use case.
Brief Description:	A strict, internal compliance check that cross-references patient identifiers against consent registries to approve or block the processing of health information.
Actors:	Primary: System
Related Use Cases:	Translate Data to/from PH Core HL7 FHIR R4/HL7 v2 (<<include>>)
Stakeholders:	Patients: Require strict adherence to their privacy choices. WAH4PCE Admin: Needs a reliable audit trail of compliance checks.
Preconditions:	Patient identifiers must be parsable from the payload, and the consent registry must be online.
Postconditions:	A consent status is appended to the transaction, and the verification event is logged.
Flow of Activities:	System: Extracts patient identifiers from the active payload. System: Queries the consent registry for the patient's sharing permissions. System: Evaluates permissions against the current transaction. System: Returns an authorization approval back to the translation process.
Exception Conditions:	E1: Consent Revoked. If the registry indicates an opt-out, the System immediately purges the payload, aborts processing, and logs a compliance rejection.